Practical 1: Installation of Hadoop Framework, its components, and study of the HADOOP ecosystem.

Part 1: Study of the Hadoop Ecosystem

Hadoop is not just one tool; it is a framework of open-source utilities.

- **HDFS (Hadoop Distributed File System):** The storage layer. It breaks large files into blocks and distributes them across nodes in a cluster.
- **YARN (Yet Another Resource Negotiator):** The resource management layer. It allocates system resources (RAM, CPU) to the various applications running in a Hadoop cluster.
- **MapReduce:** The original processing engine. It splits processing tasks into a "Map" phase (filtering/sorting) and a "Reduce" phase (summarizing). We will use this in Practicals 3, 4, 8, and 9.
- **Ecosystem Extensions:** In later practicals, you will also use **Flume** (for ingesting streaming data like Twitter), **HBase** (a NoSQL database built on top of HDFS), and **MongoDB** (a document-based NoSQL database).

Part 2: Installation of Hadoop on Windows 11

Step 1: Install Java 8 (Crucial Requirement)

Hadoop heavily relies on Java, and specifically, it is most stable with **Java 8 (JDK 1.8)**. Newer versions of Java will cause compatibility errors.

1. Download Java JDK 8 from the Oracle website.
2. **Important:** Install it in a directory with *no spaces* in the path. For example, install it directly to `C:\Java\jdk1.8.0_xx` instead of `C:\Program Files\Java\...`.

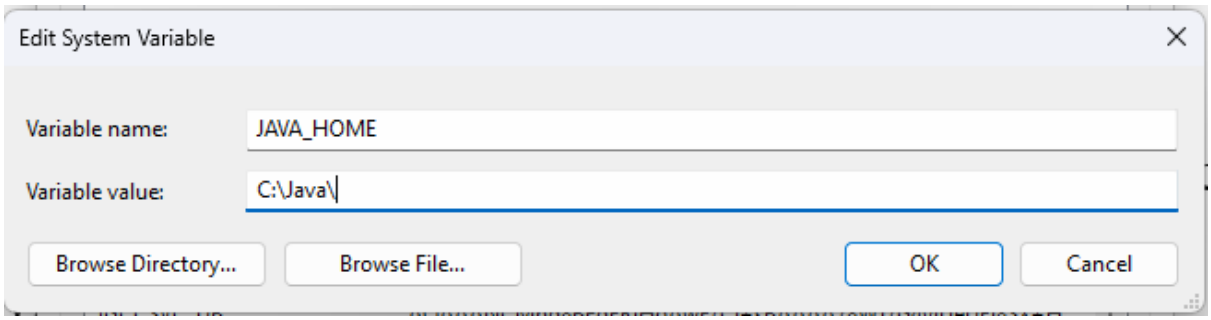
Step 2: Download and Extract Hadoop

1. Go to the official Apache Hadoop releases page.
2. Download a stable binary release (e.g., **Hadoop 3.3.6** `tar.gz` file).
3. Extract the downloaded file using a tool like 7-Zip or WinRAR.
4. Move the extracted folder directly to your C: drive so the path is clean, e.g., `C:\hadoop`.

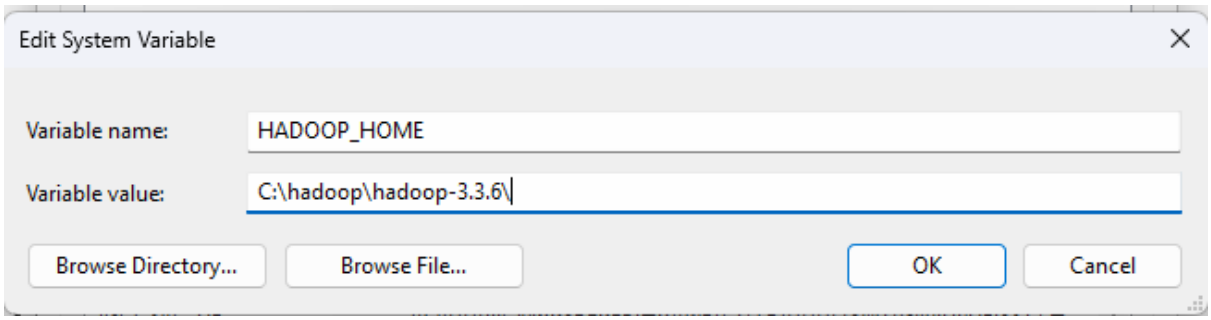
Step 3: Set Environment Variables

Windows needs to know where Java and Hadoop live.

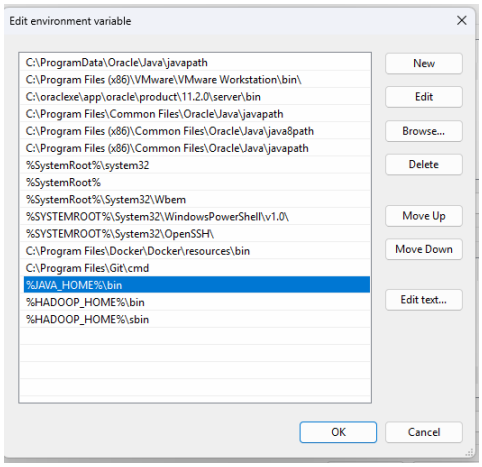
1. Open Windows Search and type **"Environment Variables"** -> click **"Edit the system environment variables"**.
2. Click the **Environment Variables** button.
3. Under **System variables**, click **New**:
   - Variable name: `JAVA_HOME`
   - Variable value: `C:\Java\jdk1.8.0_xx` (your exact Java path)



4. Click **New** again:
   - Variable name: `HADOOP_HOME`
   - Variable value: `C:\hadoop`



5. Find the `Path` variable in the list, select it, and click **Edit**. Add two new entries:
   - `%JAVA_HOME%\bin`
   - `%HADOOP_HOME%\bin`
   - `%HADOOP_HOME%\sbin`

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Step 4: Configure Hadoop XML Files

Navigate to `C:\hadoop\etc\hadoop`. You need to edit four files using Notepad.

1. core-site.xml (Place inside the `<configuration>` tags)
XML

```
<property>
   <name>fs.defaultFS</name>
   <value>hdfs://localhost:9000</value>
</property>
```

2. hdfs-site.xml

First, create a new folder `C:\hadoop\data`. Inside it, create two folders: `namenode` and `datanode`. Then, add this to the XML:

```
<property>
   <name>dfs.replication</name>
   <value>1</value>
</property>
<property>
   <name>dfs.namenode.name.dir</name>
   <value>/C:/hadoop/data/namenode</value>
</property>
<property>
   <name>dfs.datanode.data.dir</name>
   <value>/C:/hadoop/data/datanode</value>
</property>
```

3. mapred-site.xml

```
<property>
   <name>mapreduce.framework.name</name>
   <value>yarn</value>
```

```
</property>
```

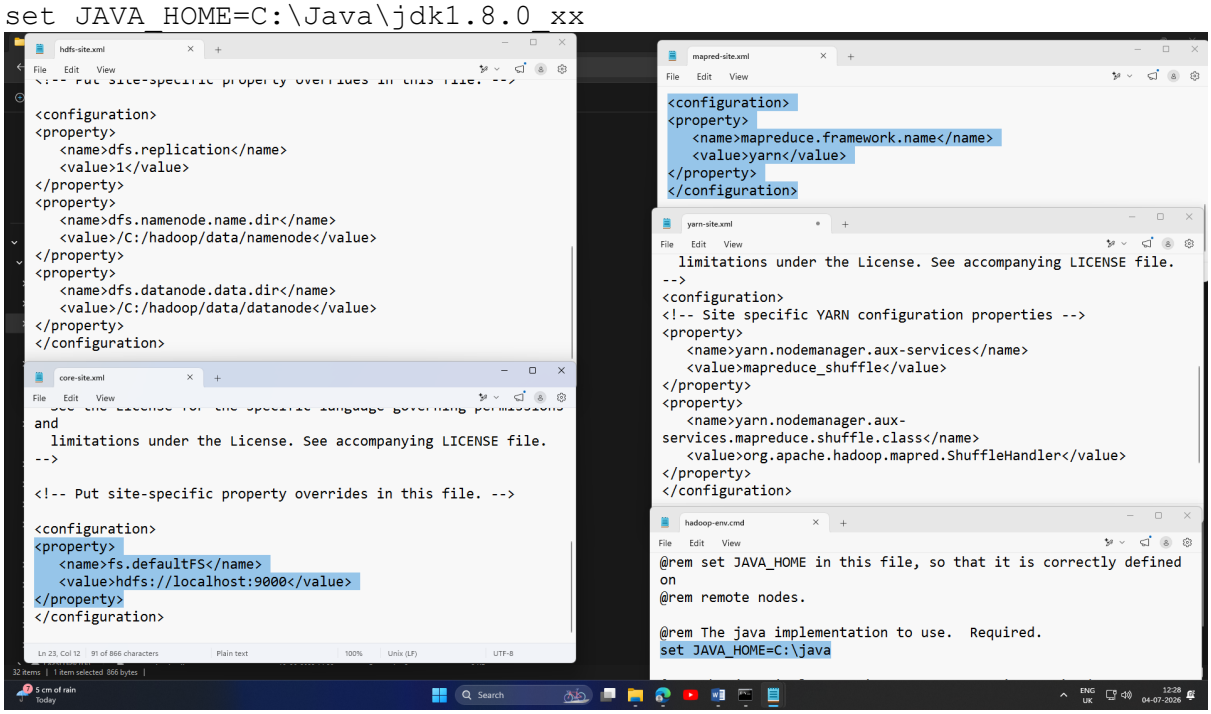
4. yarn-site.xml

```
<property>
   <name>yarn.nodemanager.aux-services</name>
   <value>mapreduce_shuffle</value>
</property>
<property>
   <name>yarn.nodemanager.aux-
services.mapreduce.shuffle.class</name>
   <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
```

5. hadoop-env.cmd

Find the line that says `set JAVA_HOME=%JAVA_HOME%` and change it to point explicitly to your Java path:

```
set JAVA_HOME=C:\Java\jdk1.8.0_xx
```

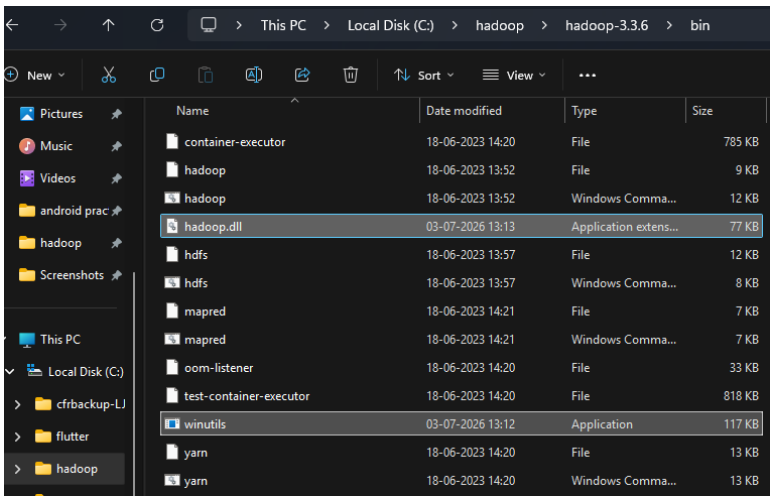


Step 5: The "Winutils" Patch

Since we are on Windows, we need Windows-specific execution files.

1. Search GitHub for "hadoop winutils" (or go to cdarlint/winutils repository).
2. Find the folder matching your Hadoop version (e.g., `hadoop-3.3.6/bin`).
3. Download the `winutils.exe` and `hadoop.dll` files.

4. Paste these files directly into your `C:\hadoop\bin` folder, replacing any existing files.



Step 6: Format and Start

1. Open Command Prompt **as Administrator**.
2. Format the Hadoop filesystem (you only do this once!):

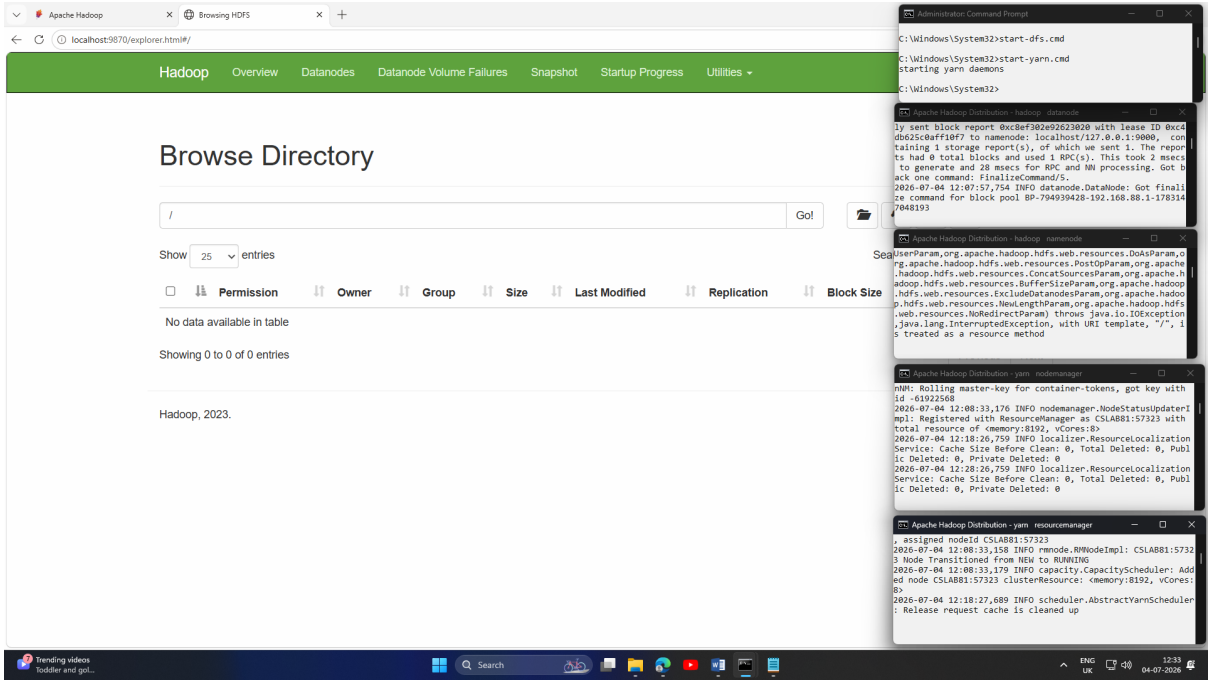
```
hdfs namenode -format
```

3. Once formatting is successful, start the Hadoop daemons:

```
start-dfs.cmd
start-yarn.cmd
```

4. **Four new command prompt windows should open** (Namenode, Datanode, ResourceManager, NodeManager).

You can verify everything is working by opening your browser and going to `http://localhost:9870` to see the Hadoop dashboard.


Apache Hadoop
Browsing HDFS
localhost:9870/explorer.html#/
Hadoop
Overview
Datanodes
Datanode Volume Failures
Snapshot
Startup Progress
Utilities
Browse Directory
/
Go!
Show 25 entries
Permission
Owner
Group
Size
Last Modified
Replication
Block Size
No data available in table
Showing 0 to 0 of 0 entries
Hadoop, 2023.
Administrator: Command Prompt
C:\Windows\System32>start-dfs.cmd
C:\Windows\System32>start-yarn.cmd
starting yarn daemons
C:\Windows\System32>
Apache Hadoop Distribution - hadoop datanode
Apache Hadoop Distribution - hadoop namenode
Apache Hadoop Distribution - yarn nodemanager
Apache Hadoop Distribution - yarn resourcemanager
Search
ENG
UK
12:33
04-07-2026